Specialist Course: Lattice Field Theory
Final Project: Comparison of Algorithms

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Scalar ϕ^4 Theory

Scalar ϕ^4 theory is the simplest relativistic system in which we can see spontaneous symmetry breaking of discrete symmetries ($\mathbb{Z}_2$ reflection symmetry, $\phi(x) \to -\phi(x)$). The difference between explicit and spontaneous breaking of a symmetry is that with spontaneous symmetry breaking, the Lagrangian is invariant under the symmetry, but the gorund state is not. In explicit symmetry breaking, the symmetry was never there to begin with. In spontaneous symmetry breaking, a phase transition occurs. For discrete symmetries, spontaneous symmetry breaking looks very similar to explicit symmetry breaking. Breaking of a continous global symmetry (for example a global $U(1)$ phase symmetry) would lead to massless particles (Goldstone theorem) and for gauged symmetries, this could explain the Higgs mechanism. However, this is beyond the scope of this project. This section is provided for completeness and is heavily based on the tutorial of the LFT course and the following book: [5].

2.1 The ϕ^4 Theory Action

We work in two-dimensional Euclidian space. Then, the ϕ^4- theory action is described by the following Lagrangian:

$$S[\phi] = \int d^2x \left[\frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) + \frac{1}{2}m^2\phi^2 + \frac{1}{4!}\lambda\phi^4\right], \tag{2.1.1}$$

where m^2 and λ are the parameters of the theory. We could regard these parameters as being temperature dependent, then this model could describe a system such as a ferromagnet near its critical temperature.
We assume periodic boundary conditions in all dimensions, hence by integrating by parts, we obtain

$$S[\phi] = \int \mathrm{d}^2x \,\left[-\frac{1}{2}\phi\,(\partial_\mu\partial^\mu\phi) + \frac{1}{2}m^2\phi^2 + \frac{1}{4!}\lambda\phi^4\right]. \tag{2.1.2}$$

The total derivative term vanished because of the periodic boundary conditions. This rewriting shows explicitly that the kinetic energy term becomes a nearest neighbour coupling when discretizing the action.

2.2 Different Masses m^2

The action has the following scalar potential:

$$V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{1}{4!}\lambda\phi^4. \tag{2.2.1}$$

The parameter λ has the same dimension as a squared mass. We assume that $\lambda > 0$, such that the potential is bounded from below. This is required to define a proper distribution, as it ensures that the Boltzmann weight is normalizable. This potential can model a system near a phase transition, such as a ferromagnet near the critical temperature T_c. The mass parameter m^2 plays a role analogous to the inverse correlation length, its sign determines the phase of the system. We can imagine the mass to be temperature dependent:

$$m^2(T) = c(T - T_c), \tag{2.2.2}$$

where c is some constant [5]. We distinguish for the following two cases for the mass parameter:

- If $m^2 \geq 0$ ($T \geq T_c$), the potential has a minimum at $\phi = 0$, which corresponds to a minimum of the action. There is a single vacuum state for which $\langle\phi\rangle = 0$.
- If $m^2 < 0$ ($T < T_c$), $\phi = 0$ is a local maximum and there are two global minima at $\phi = \pm\sqrt{\frac{-6m^2}{\lambda}}$. There are now two vacuum states and $\langle\phi\rangle = \pm\sqrt{-\frac{6m^2}{\lambda}}$.

Since the vacuum expectation value $\langle\phi\rangle = 0$ is invariant under $\phi \to -\phi$, but the two vacua $\langle\phi\rangle = \pm\sqrt{-6m^2/\lambda}$ are not, this is a case of spontaneous symmetry breaking. That means there occurs a phase transition at some critical mass value $m_c^2 < 0$. At this critical point, the correlation length diverges. This is the point where Monte-Carlo simulations become critical-slowing-down limited.

In this picture, spontaneous symmetry breaking is associated with a continuous phase transition where the effective mass crosses zero. The order parameter $\langle\phi\rangle$ changes from zero in the symmetric phase to a nonzero value in the broken phase, analogous to the magnetization of a ferromagnet emerging below the Curie temperature.

In Euclidian lattice simulations, the temperature does not appear explictly (where for example in the Ising model, it does). Changing λ and m^2 plays the role of changing extrenal thermodynamical variables. This also shows that the scalar ϕ^4 theory with its spontaneous $\mathbb{Z}_2$ symmetry breaking mirrors the magnetization transition of the Ising model. For more details about the Ising model, see [7].

2.3 Discretization of Scalar ϕ^4 Theory

This section is based on the tutorial of the LFT course.
We discretize this theory on a two-dimensional spacetime lattice of L^2 lattice sites with lattice spacing a and denote the spacetime points as

$$\vec{x} = a\vec{n} = a(n_1, n_2), \quad 0 \leq n_1, n_2 < L. \tag{2.3.1}$$

We approximate the integral as a sum over all lattice sites:

$$\int \mathrm{d}^2x \to a^2 \sum_{\vec{n}}, \tag{2.3.2}$$

For the discretization of the second derivative, we use the five-point stencil discretization (more on that, see [8]):

$$\begin{aligned}\partial_\mu\partial^\mu\phi(\vec{x}) &= \left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2}\right)\phi(\vec{x}) \\ &\to \frac{\phi(n_1+1,n_2) + \phi(n_1-1,n_2) + \phi(n_1,n_2+1) + \phi(n_1,n_2-1) - 4\phi(\vec{n})}{a^2} + \mathcal{O}(a^2) \\ &= \sum_{\mu\in\{1,2\}} \frac{1}{a^2}\left[\phi(\vec{n}+\hat{\mu}) + \phi(\vec{n}-\hat{\mu}) - 2\phi(\vec{n})\right] + \mathcal{O}(a^2).\end{aligned} \tag{2.3.3}$$

All together, the discretized action reads

$$S[\phi] = a^2\sum_{\vec{n}}\left(-\frac{1}{2a^2}\phi(\vec{n})\sum_{\mu}\left[\phi(\vec{n}+\hat{\mu}) + \phi(\vec{n}-\hat{\mu}) - 2\phi(\vec{n})\right] + \frac{1}{2}m^2\phi^2(\vec{n}) + \frac{1}{4!}\lambda\phi^4(\vec{n})\right) \tag{2.3.4}$$

or, after some algebra,

$$\boxed{S[\phi] = \sum_{\vec{n}}\left(\left[2 + \frac{1}{2}m^2\right]\phi^2(\vec{n}) + \frac{1}{4!}\lambda\phi^4(\vec{n}) - \sum_{\mu}\left[\phi(\vec{n}+\hat{\mu})\phi(\vec{n})\right]\right)} \tag{2.3.5}$$

where we have redefined $(am)^2 \to m^2$ and $(a^2\lambda) \to \lambda$. The discretized action has a changed mass, $m_{\text{eff}}^2 = 2 + \frac{1}{2}m^2$.

Chapter 3

Algorithms

3.1 General Introduction to Monte Carlo Methods

Our goal is to evaluate the expectation value of an observable $O[U]$ in the qunatizes Euclidian gauge field theory on a lattice. This is written as a path integral (functional)

$$\langle O \rangle = \frac{1}{Z} \int \mathcal{D}[U] \; \exp\left(-S[U]\right) O[U], \tag{3.1.1}$$

where the partition function is defined as

$$Z = \int \mathcal{D}[U] \exp\left(-S[U]\right). \tag{3.1.2}$$

In the general case, these integrals cannot be solved analytically and depending on the application, these can be high-dimensional [3].
Using probability theory, we can approximate the integral of a function $f(x)$ by averaging the functional values $f(x_n)$ at values x_n, which are randomly chosen according to a uniform distribution

$$\rho_u(x_n) = \frac{1}{b-a}. \tag{3.1.3}$$

The integral is approximated as:

$$\frac{1}{b-a} \int_a^b dx \; f(x) = \lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f(x_n). \tag{3.1.4}$$

The exact mean is replaced by the sample mean, the calculation is restricted to a subsample and a finite number N. By the Strong Law of Large Numbers, the Monte Carlo estimator

$$\frac{1}{N} \sum_{n=1}^{N} f(x_n)$$

converges almost surely to the exact expectation value as $N \to \infty$, and the Central Limit Theorem implies an error scaling of $\mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$. This means to increase the accuracy by a factor of two, one has to use four times as many samples [3].

3.1.1 Example of a Simple Monte Carlo Integral

Suppose we want to estimate the integral

$$\int_0^1 \exp(-x^2) \, dx. \tag{3.1.5}$$

We restrict the upper limit to 1 to match the support of a uniform random variable. Let X be a random variable uniformly distributed on the interval $(0,1)$. Its probability density function (pdf) is given by

$$f(x) = \begin{cases} 1 & \text{if } x \in (0,1), \\ 0 & \text{otherwise.} \end{cases} \tag{3.1.6}$$

Then the integral can be written as an expectation,

$$\int_0^1 \exp(-x^2)\, dx = \mathbb{E}[\exp(-X^2)]. \tag{3.1.7}$$

A Monte Carlo estimator using N independent samples $X_1, X_2, \ldots, X_N$ from the uniform distribution is

$$I_N = \frac{1}{N} \sum_{i=1}^{N} \exp(-X_i^2). \tag{3.1.8}$$

This estimator converges to the true value as $N \to \infty$,

$$I_N \xrightarrow{N\to\infty} \int_0^1 \exp(-x^2)\, dx.$$

This simple example illustrates how Monte Carlo methods approximate integrals by replacing them with sample averages [1].

3.1.2 Importance Sampling

In our path integral in Equation 3.1.1, we need to account for the Boltzmann factor $\exp(-S)$. This is a weighting factor, depending on the action S, of the different field configurations. The space of all configurations is not uniformely explored, but according to the weights determined by the action S. The main idea of importance sampling is to approximate a large sum with a comparatively small amount of configurations, sampled according to this weight factor.
For a general probability distribution $\rho(x)$, the expectation value of a function $f(x)$ is

$$\langle f \rangle_\rho = \frac{\int_a^b dx\ \rho(x) f(x)}{\int_a^b dx\ f(x)}. \tag{3.1.9}$$

In the importance sampling Monte Carlo integration, this is approximated as follows:

$$\langle f \rangle_\rho = \lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f(x_n), \tag{3.1.10}$$

where $x \in (a,b)$ sampled according to the normalized probability density $\frac{\rho(x)dx}{\int_a^b dx \rho(x)}$. Our path integral (Equation 3.1.1) is of this form, so we can apply this method:

$$\langle O \rangle = \lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} O[U_n], \tag{3.1.11}$$

with U_n sampled according to the probability distribution density

$$dP(U) = \frac{e^{-S[U]}\mathcal{D}[U]}{\int \mathcal{D}[U] e^{-S[U]}}, \tag{3.1.12}$$

which is called the Gibbs-measure [3].

3.1.3 Markov Chains

The configurations U_n are generated in a sequence, a so-called homogeneous Markov chain: $U_0 \to U_1 \to U_2 \to \dots$. The first configuration U_0 is chosen randomly, the update from one configurations to the next is called the Markov step. The next configuration only depends on the present one, not one the history of the process. This condition is written as

$$P(U_n = U'|U_{n-1} = U) = T(U'|U). \tag{3.1.13}$$

The configuration sequence explores the space of all configurations according to the probability density defined by the action. Eventually, the Markov chain will reach an equilibrium, after that, it cannot have sinks or sources of probability. This gives rise to a balancing condition:

$$\sum_U T(U|U')P(U) = \sum_U T(U'|U)P(U'). \tag{3.1.14}$$

A solution for the balancing condition in Equation 3.1.14 can be obtained by requiring term-wise equality:

$$T(U|U')P(U) = T(U'|U)P(U'). \tag{3.1.15}$$

This is called detailed balance condition, it is the most commonly used solution.
Using the normalization property, we find in addition that

$$P(U') = \sum_U T(U|U')P(U). \tag{3.1.16}$$

The equilibrium distribution $P(U)$ is a fixed point of the Markov process, once the equilibrium is reached, the system stays there forever.
In order to give correct result, the Markov chain must be aperiodic, i.e. be able to reach all configurations in the configuration space. This is the case iff the transition matrix $T(U|U')$ is positive-semidefinite. This property is called strong ergodicity [3].

3.2 The Metropolis Algorithm

The Metropolis Algorithm is used to obtain a new configuration U_{n+1} from the current configuration U_n. We use $P[U] \propto \exp(S[U])$. The steps are as follows:

1. Choose a candidate configuration U' according to some a priori selection probability $T_0(U'|U)$, where $U = U_n$.
2. Accept the candidate configuration U' as the new configuration U_{n+1} with the following probability:

$$T_A(U'|U) = \min\left(1, \frac{T_0(U|U')\exp(-S[U'])}{T_0(U'|U)\exp(-S[U])}\right). \tag{3.2.1}$$

   If the change is rejected, keep the old configuration, set $U_{n+1} = U_n$.

3. Repeat.

The detailed balance condition is fulfilled:

$$\begin{aligned}
T(U'|U)\exp(-S[U]) &= T_0(U'|U)\min\left(1, \frac{T_0(U|U')\exp(-S[U'])}{T_0(U'|U)\exp(-S[U])}\right)\exp(-S[U]) && (3.2.2)\\
&= \min\left(T_0(U|U')\exp(-S[U']), T_0(U'|U)\exp(-S[U])\right) && (3.2.3)\\
&= T(U|U')\exp(-S[U']) && (3.2.4)
\end{aligned}$$

We have used the symmetry of min and the positivity of all factors.

Often, we have $T_0(U|U') = T_0(U'|U)$. In this case, the transition probability simplifies to

$$T_A(U'|U) = \min\left(1, \exp(-\Delta S)\right), \quad \text{with} \quad \Delta S = S[U'] - S[U]. \tag{3.2.5}$$

The decision to accept or reject only depends on the change in the action between the two configurations. Any normalization constant in $P[U]$ will cancel automatically, note that this is crucial in field theory where the partition function Z is unknown. In practice, the acceptance rate is strongly dependent on the proposed change. Very large proposed updates are often rejected, very small ones lead to long autocorrelation times. Hence we often aim for acceptance rates between 50% - 80%.

A warm-up period (thermalization) is required to eliminate dependence on the initial configuration. After that, configurations are still correlated, and one must estimate autocorrelation times when computing errors.

It can be shown that the Metropolis algorithm fulfills ergodicity, meaning that any configuration is reachable from any other configuration with positive probability in a finite number of steps. Because the transition kernel satisfies detailed balance with respect to $\exp(-S[U])$, this probability distribution is the stationary distribution of the Markov chain. Together with ergodicity, this ensures convergence of the algorithm to the desired ensemble.

3.3 The Hybrid Monte Carlo Algorithm (HMC)

The Hybrid Monte Carlo Algorithm (HMC) algorithm was first introduced in this paper: [2]. The authors solved the major computational challenge of involving fermions into lattice field theory.

Simulating fermions in a lattice field theory is difficult, since fermions are described by Grassmann numbers, for which Monte Carlo methods cannot be applied directly. Integrating over fermions produces very non-local effects in the action, which makes calculations extremely expensive. To solve this problem, the nested Monte Carlo steps are replaced with an equation of motion in a fictitious time variable. These are solved numerically with different techniques. Evolving the entire field configuration simultaneously using a numerical integrator avoids the locality problem but introduces truncation errors from discrete integration. Although these errors can be extrapolated away, they remain undesirable. The paper [2] introduces the HMC, which addresses the problem with the following steps:

- evolve all fields in parallel using a molecular-dynamics-like update,
- a global Metropolis accept/reject step, which removes the truncation errors.

This method combines deterministic molecular dynamics and accept/reject steps, achieving efficiency and exactness. Even though this algorithm was originally developed for lattice field theories, it is nowadays applied in various other fields.

The goal is again to calculate the expectation of some operator along with its partition function, as defined in the Equations 3.1.2 and 3.1.2. The Metropolis algorithm is a rater inefficient way to do this and we search for a method to choose candidate configurations efficiently. We aim for a high acceptance rate and minimum correlation between successive configurations. For this, we introduce an artificial computer time τ and a hamiltonian dynamics to describe the developement of the field $\phi(\tau)$. In addition, we will need the conjugate momenta $\pi(\tau)$. This means, instead of sampling the distribution $p(\phi) = \frac{1}{Z}e^{-S(\phi)}$, we sample $p(\phi, \pi) = \frac{1}{Z'}e^{-H[\phi,\pi]}$. The Hamiltonian $H[\phi, \pi]$ is given as follows:

$$H[\pi, \phi] = \frac{1}{2}\pi^2 + S(\phi), \tag{3.3.1}$$

where $S(\phi)$ is an arbitrary action. A key point of this algorithm is that the expectation values over $p(\phi)$ are the same as those over $p(\phi, \pi)$ up to some constant, since the integration over $\mathcal{D}\pi$ is a Gaussian integral, which gives rise to a constant. This constant will cancel out in the accept/reject step.

Introducing the conjugate momenta translates the numerical evaluation of the path integral into the language of classical Hamiltonian mechanics. The HMC algorithm solves the classical equations of motion for each lattice site x:

$$\dot{\phi}_x = \frac{d\phi_x}{d\tau} = \frac{\partial H}{\partial \pi} = \pi_x \tag{3.3.2}$$

$$\dot{\pi}_x = \frac{d\pi_x}{d\tau} = -\frac{\partial H}{\partial \phi_x} = -\frac{\partial S}{\partial \phi_x} = -F[\phi_x] \qquad \text{"force"} \tag{3.3.3}$$

Because of the conservation of energy, the Hamiltonian remains constant. This can be used to check the accuracy of the solutions of the differential equations, which have to be solved numerically. In Chapter 4, there will be a description of different methods to achieve this numerical solution.

The procedure for generating a new configuration ϕ' is to select some initial momenta π from a standard Gaussian distribution (zero mean, unit variance)

$$P_G(\pi) \propto \exp\left(-\frac{\pi^2}{2}\right), \tag{3.3.4}$$

and then evolve the system through the (ϕ, π)- phase space for a time τ_0 according to Hamilton's equations 3.3.2. The probability of choosing the configuration (ϕ', π') is

$$P_H((\phi,\pi) \to (\phi',\pi')) = \delta\left[(\phi',\pi') - (\phi(\tau_0), \pi(\tau_0))\right]. \tag{3.3.5}$$

The acceptance probability for this candidate is

$$P_A((\phi,\pi) \to (\phi',\pi')) = \min\left(1, \exp(\delta H)\right), \tag{3.3.6}$$

where $\delta H = H(\phi', \pi') - H(\phi, \pi)$.

In order for the HMC to correctly sample the target distribution $p(\phi) \propto e^{-S(\phi)}$, the transition probability restricted to the field variables ϕ alone must satisfy detailed balance. Starting from the full transition probability in (ϕ, π) space, the transition probability for $\phi \to \phi'$ is obtained by integrating out the momenta:

$$P_M(\phi \to \phi') = \int [d\pi]\,[d\pi']\; P_G(\pi)\, P_H\big((\phi,\pi) \to (\phi',\pi')\big)\, P_A\big((\phi,\pi) \to (\phi',\pi')\big). \tag{3.3.7}$$

Here $P_G(\pi)$ is the Gaussian distribution used to refresh the momenta, P_H is the deterministic Hamiltonian evolution, and P_A is the Metropolis accept/reject step.

The main requirement for detailed balance is that the molecular-dynamics evolution is reversible. This means that evolving (ϕ, π) forward in fictitious time τ to (ϕ', π') is equivalent to evolving $(\phi', -\pi')$ backward to $(\phi, -\pi)$:

$$P_H\big((\phi,\pi) \to (\phi',\pi')\big) = P_H\big((\phi',-\pi') \to (\phi,-\pi)\big). \tag{3.3.8}$$

This property holds automatically for Hamiltonian dynamics because the flow generated by a Hamiltonian is time-reversal invariant when accompanied by a sign flip of all momenta. From the following identity (which we already used to show detailed balance for Metropolis)

$$e^{-H(\phi,\pi)} \min(1, e^{-\Delta H}) = \min\Big(e^{-H(\phi,\pi)}, e^{-H(\phi',\pi')}\Big) \tag{3.3.9}$$

$$= e^{-H(\phi',\pi')} \min(1, e^{+\Delta H}), \tag{3.3.10}$$

the fact that the Hamiltonian $H(\phi, \pi)$ is also invariant under $\pi \to -\pi$ and $P_s(\phi)P_G(\pi) \propto \exp(H(\phi, \pi))$, we obtain

$$P_S(\phi)\, P_G(\pi)\, P_A((\phi,\pi) \to (\phi',\pi')) = P_S(\phi')\, P_G(\pi')\, P_A((\phi',\pi') \to (\phi,\pi)). \tag{3.3.11}$$

Multiplying this by the transition probability P_H and integrating over π, π' gives

$$\int [d\pi]\,[d\pi']\; P_S(\phi)\, P_G(\pi)\, P_H((\phi,\pi) \to (\phi',\pi'))\, P_A((\phi,\pi) \to (\phi',\pi')) \tag{3.3.12}$$

$$= \int [d\pi]\,[d\pi']\; P_S(\phi')\, P_G(\pi')\, P_H((\phi',\pi') \to (\phi,\pi))\, P_A((\phi',\pi') \to (\phi,\pi)). \tag{3.3.13}$$

Because the measure is invariant under momentum reversal,

$$[d\pi]\,[d\pi'] = [d(-\pi)]\,[d(-\pi')],$$

and using the reversibility condition from above, this reduces precisely to

$$P_S(\phi)\, P_M(\phi \to \phi') = P_S(\phi')\, P_M(\phi' \to \phi), \tag{3.3.14}$$

which is the detailed balance condition for P_M.

By conservation of energy, we need $H = H'$, meaning $\delta H = 0$ and $P_A = 1$. However, solving differential equations numerically always interoduces errors, hence in general, $P_A \leq 1$. We have just proved that the HMC produces ϕ configurations with the correct distribution for any $\delta H \neq 0$. Also note that the momenta π are not used to compute the observables. They can be disregareded in each step [2].

Note that the accept/reject step plays a crucial role in the HMC. Hamiltonian evolution exchanges energy between the kinetic term $\frac{1}{2}\pi^2$ and the potential term $S(\phi)$ while (ideally) keeping their sum $H = \frac{1}{2}\pi^2 + S(\phi)$ constant. The accept/reject step corrects the small energy violations $\delta H \neq 0$ resulting from the numerical integration, as it compares the Hamiltonians and accepts the proposed configuration with probability P_A. This enforces the correct equilibrium distribution, as a substantial increase in the energy will be rejected. In addition, the accept/reject step also ensures ergodicity. Pure Hamiltonian evolution is deterministic and reversible, and without momentum refreshment or a mechanism to leave a given energy surface, the algorithm would explore only a single trajectory in phase space. Periodically resampling the momenta from a Gaussian distribution redistributes kinetic energy, allowing the system to have different energies [2].

Chapter 4

Integrators for HMC

Our aim is to solve the Hamiltonian equations of motion 3.3.2, which are at the heart of the HMC algorithm, numerically. This Chapter describes different methods used for this task.

4.1 Leapfrog Integration

In Section 3.3, we have discussed that the detailed balance condition for the HMC algorithm imposes the following two requirements on the molecular dynamics trajectory:

- Invariance of the integration measure under Hamiltonian evolution in the (ϕ, π) phase space:

$$[\mathcal{D}\phi][\mathcal{D}\pi] = [\mathcal{D}\phi(\tau)][\mathcal{D}\pi(\tau)] \tag{4.1.1}$$

- Reversibility of the trajectory, evolving forward in fictitious time τ and then reversing the momenta exactly retraces the trajectory

The leapfrog integration method fulfills these requirements, as will be shown in the following. Leapfrog is a second-order method, compared to the very-well known Euler method.
In the leapfrog integration method, the field ϕ is evolved in n steps of length ε. The conjugate momenta start with a half-step of $\frac{\varepsilon}{2}$, then $(n-1)$ full steps and finally other half-step. A single such step $(n = 1)$ is denoted as follows:

$$\phi(0) \to \phi(\varepsilon) \tag{4.1.2}$$

$$\pi(0) \to \pi(\frac{\varepsilon}{2}) \to \pi(\varepsilon). \tag{4.1.3}$$

The simplest integration scheme is second-order accurate:

$$\phi(0) \to \phi(\varepsilon) = \phi(0) + \pi(\frac{\varepsilon}{2})\varepsilon \tag{4.1.4}$$

$$\pi(0) \to \pi(\frac{\varepsilon}{2}) = \pi(0) - \frac{\partial S}{\partial \phi}|_{\phi(0)}\frac{\varepsilon}{2} \to \pi(\varepsilon) = \pi(\frac{\varepsilon}{2}) - \frac{\partial S}{\partial \phi}|_{\phi(\varepsilon)}\frac{\varepsilon}{2} \tag{4.1.5}$$

When dealing with a series of steps of length n, we use the abbreviations $\phi(n\varepsilon) = \phi_n$ and $\pi(n\varepsilon) = \pi_n$.

The invariance of the integration measure can be shown from the product of Jacobians along the sequence:

$$\det\left[\frac{\partial(\pi_1, \phi_1)}{\partial(\pi_0, \phi_0)}\right] = \det\left[\frac{\partial(\pi_1, \phi_1)}{\partial(\pi_{\frac{1}{2}}, \phi_1)}\frac{\partial(\pi_{\frac{1}{2}}, \phi_1)}{\partial(\pi_{\frac{1}{2}}, \phi_0)}\frac{\partial(\pi_{\frac{1}{2}}, \phi_0)}{\partial(\pi_0, \phi_0)}\right] = 1 \tag{4.1.6}$$

This factorizes into a product of three determinants of upper diagonal matrices with 1 on the diagpnal, hence the result equals 1 and the integration measure is invariant.

For the reversibility property, use the scheme in Equations 4.1.4 starting from (ϕ_0, π_0):

$$\phi_1 = \phi_0 + \pi_0\varepsilon - \frac{1}{2}\frac{\partial S}{\partial \phi}|_{\phi_0}\varepsilon^2 \tag{4.1.7}$$

$$\pi_1 = \pi_0 - \frac{1}{2}\left(\frac{\partial S}{\partial \phi}|_{\phi_0} + \frac{\partial S}{\partial \phi}|_{\phi_1}\right)\varepsilon \tag{4.1.8}$$

To go backwards, we start at (ϕ_1, π_1) and apply the equations of motion 4.1.4 with a negative step size $-\varepsilon$. This gives

$$\begin{aligned}
\phi(\varepsilon - \varepsilon) &= \phi_1 - \pi_1\varepsilon - \frac{1}{2}\frac{\partial S}{\partial \phi}|_{\phi_1}\varepsilon^2 && (4.1.9)\\
&= \phi_1 - \pi_1\varepsilon + \frac{1}{2}\left(\frac{\partial S}{\partial \phi}|_{\phi_0} + \frac{\partial S}{\partial \phi}|_{\phi_1}\right)\varepsilon^2 - \frac{1}{2}\frac{\partial S}{\partial \phi}|_{\phi_1}\varepsilon^2 && (4.1.10)\\
&= \phi_1 - \pi_1\varepsilon + \frac{1}{2}\frac{\partial S}{\partial \phi}|_{\phi_0}\varepsilon^2 = \phi_0, && (4.1.11)
\end{aligned}$$

$$\pi(\varepsilon - \varepsilon) = \pi_0 = \pi_1 + \left(\frac{\partial S}{\partial \phi}|_{\phi_1} + \frac{1}{2}\frac{\partial S}{\partial \phi}|_{\phi_0}\right) = \pi_0. \tag{4.1.12}$$

So the reversibility is proven. It is possible to combine n leapfrog steps. The discretization error for half-steps is $\mathcal{O}(\varepsilon^2)$, for full steps $\mathcal{O}(\varepsilon^3)$ [3].

4.2 The Omelyan2 Integrator

This integrator is described in great detail in the following papers: [6, 4].

The Omelyan2 integrator is a generalization of the standard leapfrog scheme, designed to improve energy conservation while preserving the two key properties required for HMC:

- Invariance of the integration measure in the (ϕ, π) phase space
- Reversibility of the trajectory under time reversal

Like leapfrog, Omelyan integration evolves the fields ϕ and conjugate momenta π in a sequence of small steps. The key difference is that the momentum and position updates are distributed asymmetrically within each step according to a free parameter λ, which can be tuned to minimize the discretization error of the Hamiltonian.
A single Omelyan step of size ε can be written as:

$$\pi \to \pi - \lambda\frac{\varepsilon}{2}\frac{\partial S}{\partial \phi} \tag{4.2.1}$$

$$\phi \to \phi + \frac{\varepsilon}{2}\pi \tag{4.2.2}$$

$$\pi \to \pi - (1-2\lambda)\frac{\varepsilon}{2}\frac{\partial S}{\partial \phi} \tag{4.2.3}$$

$$\phi \to \phi + \frac{\varepsilon}{2}\pi \tag{4.2.4}$$

$$\pi \to \pi - \lambda\frac{\varepsilon}{2}\frac{\partial S}{\partial \phi} \tag{4.2.5}$$

Here, λ is a free parameter, and the standard leapfrog scheme is recovered for $\lambda = 0$. In practice, a common choice is $\lambda \approx 0.1931833$, which minimizes the error in the Hamiltonian for harmonic systems [6, 4].

The Omelyan 2 integrator improves on the basic leapfrog scheme in the following ways:

- Reduced energy drift: By distributing the momentum updates asymmetrically, the integrator achieves better conservation of the Hamiltonian over long trajectories, which increases the acceptance rate in HMC.

- Preservation of measure and reversibility: Like leapfrog, the Omelyan 2 integrator is reversible, ensuring that detailed balance in HMC is maintained.
- Higher efficiency: For a given step size ε, Omelyan2 typically allows longer trajectories or fewer steps while achieving the same or better acceptance, reducing computational cost.

Chapter 5

Results

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